1] Logistic Regression :-

* It is a supervised machine learning algorithm which is used to solve all the classification problem

* It predicts the probability of a categorical dependent variable, usually binary (0 or 1)

* Predicts probability (0 or 1) using a sigmoid function

Types :

(1) Binary LR : 0 or 1 (Pass / Fail)

(2) Multinomial LR : More than 2 categories (Cat / Dog / Bird)

(3) Ordinal LR : Ordered categories (Low / Medium / High)

Assumptions :

(1) Independent observations

(2) Binary dependent variable

(3) Linear relationship

(4) No outliers

Mathematical implementation :

Step 1 : Linear eq'n

$$\boxed{z = w_0 + w_1 x}$$

where w_0 - bias
w_1 - weight
x - input

Step 2 : Sigmoid function

$$\boxed{\sigma(z) = \frac{1}{1+e^{-x}}}$$

range $(0 \ \& \ 1)$

Step 3 : Prediction rule :- If $\sigma(z) \geq 0.5 \rightarrow$ class $= 1$
Else $\rightarrow$ class $= 0$

Example, Given $x = 2$, $w_0 = -1$, $w_1 - 0.8$

$$z = w_0 + w_1 x$$
$$= -1 + 0.8 \times 2 = 0.6$$

$$\sigma(0.6) = \frac{1}{1+e^{-0.8}} = 0.645$$

$$0.645 > 0.5$$

Predicted class $= 1$ //

2] Support Vector Machine (SVM) :-

SVM is a supervised ML algorithm used for classification & regression, which finds optimal hyperplane that separates data points into diff classes with max margin.

hyperplane → boundary : it is used to get max margin b/w classes based on support vectors

margin : it is used to show the distance b/w hyperplane & nearest data point

support vectors : connecting with critical data points that will be closest to the boundary

Types :- (1) Linear SVM ; • Data is linearly separable
• Use a straight line (2D) or plane

(2) Non-Linear SVM : Uses kernel trick to handle complex data

Common kernels :

- Linear kernel
- Polynomial kernel
- Radial Basis Function (RBF kernel)
- Sigmoid Kernel

Mathematical Implementation :

Step 1 : Hyperplane Equation

$$\boxed{w^T x + b = 0}$$

where w = weight Vector
x = input Vector
b = bias

Step 2 : Decision rule

If $w^T x + b \geq 0 \rightarrow$ class $+1$

If $w^T x + b < 0 \rightarrow$ class -1

Step 3 : Margin maximization $\rightarrow$ goal : maximize margin $\frac{2}{\|w\|}$

Equivalent to minimizing : $\frac{1}{2} \|w\|^2$

subject to : $y_i (w^T x_i + b) \geq 1$

Step 4 : Loss func : $L = \max(0, 1 - y(w^T x + b))$

Example : Given $x = [2, 3]$, $w = [1, -1]$, $b = -1$

$$w^T x = (1 \times 2) + (-1 \times 3) = -1$$

$$z = w^T x + b = -1 + (-1) = -2$$

$$z = -2 < 0$$

Predicted class $= -1$ //

3] K-Nearest Neighbors (KNN):

* It is a supervised ML algorithm used for classification & regression.
* Lazy learning

Mathematical Implementation:

Step 1 : Choose K (no. of neighbors)

Step 2 : Distance Calculation, most common: Euclidean distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 3 : sort distances in ascending order

Step 4 : Select K Nearest Neighbors

Step 5 : Voting :
- Classification → majority class
- Regression → avg value

Example :

Training data:

Point	X	Y	Class
A	1	1	Red
B	2	2	Red
C	3	3	Blue
D	6	6	Blue

Test point : (3,2)

Choose : $k=3$

Step 1 : $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

- Dist from A (1,1) : $d = \sqrt{(3-1)^2 + (2-1)^2} = 2.24$
- Dist from B (2,2) : $d = \sqrt{(3-2)^2 + (2-2)^2} = 1$
- Dist from C (3,3) : $d = \sqrt{(3-3)^2 + (2-3)^2} = 1$
- Dist from D (6,6) : $d = \sqrt{(3-6)^2 + (2-6)^2} = 5$

Step 2 : Sort

Point	Distance	Class
B	1	Red
C	1	Blue
A	2.24	Red
D	5	Blue

Step 3 : Top 3 nearest neighbour : B → Red
C → Blue
A → Red

Step 4 : Majority → Red = 2

Predicted class = Red

4] Naive Bayes :

It is a supervised machine learning algo based on probability theorem (Bayes theorem), used for classification tasks by assuming that features are independent of each other.

Use cases : Spam detection, Sentiment Analysis, Document Categorization, Text classification

Formula : $$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

where $P(A|B) \rightarrow$ Posterior probability

$P(B|A) \rightarrow$ Likelihood

$P(A) \rightarrow$ Prior probability

$P(B) \rightarrow$ Evidence

Types : (1) Gaussian : Used for continuous numerical data

(2) Multinomial : text classification tasks

(3) Bernoulli : word present or not

Example : Spam detection

Training data :

Message Type	Offer	Free	Class
Yes	Yes	Yes	Spam
Yes	No	Yes	Spam
No	Yes	No	Not spam
No	No	No	Not spam

$P(\text{Spam}) = \frac{2}{4} = 0.5$ $\quad$ $P(\text{Not Spam}) = \frac{2}{4} = 0.5$

$P(\text{Offer} = \text{Yes} \mid \text{Spam}) = \frac{2}{2} = 1$ $\quad$ $P(\text{Offer} = \text{Yes} \mid \text{Spam}) = \frac{2}{2}$

$P(\text{Offer} = \text{Yes} \mid \text{Not Spam}) = \frac{0}{2} = 0$ $\quad$ $P(\text{Free} = \text{Yes} \mid \text{Not Spam}) = \frac{1}{2}$

Message : Test data, offer = Yes, Free = Yes

$P(\text{Spam} \mid X) = P(\text{Spam}) \cdot P(\text{Offer} \mid \text{Spam}) \cdot P(\text{Free} \mid \text{Spam})$

$= 0.5 \times 1 \times 1 = 0.5$

$P(\text{Not Spam} \mid X) = P(\text{Not Spam}) \cdot P(\text{Offer} \mid \text{Not Spam}) \cdot P(\text{Free} \mid \text{Not}$

$= 0.5 \times 0 \times 0.5 = 0$

$0.5 > 0 \Rightarrow$ Predicted class = Spam//

5] Decision Tree :

It is a supervised ML algo, used for both classification & regression, which splits the data into smaller subsets based on decision rules.

Node → Decision point

Branch → outcome of decision

Leaf → Final prediction

Attribute Selection Measure (ASM) : Used to decide which feature to split

↳ Types :- • Information gain
• Gini index

Mathematical Implementation :

(1) Entropy (Measure of impurity)

$$\text{Entropy} = \sum p_i \log_2 (p_i)$$ where p_i = prob of class i

(2) Info Gain, IG = Entropy (parent) − Σ (weighted ent of children)

↑ Gain → Better split

(3) Gini Index, $\text{Gini} = 1 - \sum p_i^2$

↓ Gini → Better split

Example :

Outlook	Play
Sunny	No
Sunny	No
Overcast	Yes
Rain	Yes

Step 1 : Yes = 2 , No = 2

$$\text{Entropy} = -\left(\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right)$$

$$= 1$$

Step 2 :

- Sunny (2 No) => Entropy = 0
- Overcast (1 Yes) => Entropy = 0
- Rain (1 Yes) => Entropy = 0

Step 3 : Info gain, IG = 1 − 0 = 1

Step 4 : Tree

Sunny	Overcast	Rain
↓	↓	↓
No	Yes	Yes

Disadv : It cannot solve overfitting

6) Random Forest :

It is a supervised ML algo used for both classification & regression which builds multiple decision trees & combines their o/ps to improve accuracy.

Ensemble learning : combines multiple models to improve performance

Bagging (Bootstrap aggregating) :

- creates multiple datasets using random sampling with replacement
- Trains a separate decision tree on each dataset

Boosting techniques :
- Gradient Boosting
- AdaBoost
- XG Boost

From multiple models, $h_1(x), h_2(x), h_3(x) \ldots, h_N(x)$

Classification → majority, regression → avg

Ex

Tree 1 → Yes, Tree 2 → No, Tree 3 → Yes

Yes = 2, No = 1

Final = Yes //